

BEYOND MINI-BATCH RELATIONS: DISTILLING TEACHER EMBEDDING MANIFOLDS WITH MULTI-SCALE GRAPH DIFFUSION

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ABSTRACT

Compact text embedding models power semantic search, clustering, and retrieval-augmented generation, and distillation from a strong teacher is the standard way to train them. The dominant relational objective matches teacher and student similarities inside a mini-batch, but a mini-batch is a uniform sample of the corpus: an anchor’s semantic neighbors almost never appear in it, so the objective constrains only a vanishing fraction of each anchor’s local geometry and delegates the rest to the student’s own inductive bias. We propose **RIPPLE**, which replaces batch relations with anchored local supervision. RIPPLE builds a mutual k -nearest-neighbor graph over cached teacher embeddings, treats it as a discrete semantic manifold, and trains the student to match a family of graph-diffusion distributions, each at its own resolution, over sampled candidate neighborhoods—using only final teacher embeddings, with no hidden states, online teacher passes, or task labels. Our theory derives each component from the requirement it meets. Zero batch-relational loss constrains only the pairs a batch happened to contain, a fraction that provably excludes nearly all of an anchor’s neighbors at any practical schedule. Matching one-step diffusion already controls every longer diffusion time, but with an error that grows linearly in it, which is what the broader scales are for and why each must be matched at its own resolution. Diffusion-weighted sampling approximates full neighborhoods at a cost set by target sharpness rather than neighborhood size. Finally, the diffusion loss leaves exactly one similarity offset free per anchor; supervising the rows a teacher random walk visits—nodes the mini-batch has already encoded—collapses that freedom to one offset per connected component of the teacher graph. Across nine datasets and three teacher-student pairs, RIPPLE attains the best overall average among compact students in two of three settings and second best in the third, with the largest gains on semantic textual similarity.

1 INTRODUCTION

Text embedding models have become a basic infrastructure layer for semantic search, clustering, reranking, and retrieval-augmented generation. Their usefulness comes not only from individual vectors, but from the geometric structure those vectors induce: neighborhoods, semantic regions, and smooth paths between related examples. Strong recent embedding models, including LLM-based encoders, often learn rich semantic geometry but are costly to run at scale. Encoding large private collections, refreshing indexes as documents change, or serving retrieval on constrained hardware therefore motivates compact student encoders that inherit the geometry of a stronger teacher.

Knowledge distillation offers a practical route to this compression (Hinton et al., 2015). In language representation learning it has been applied through output matching, hidden-state alignment, self-attention transfer, and compact pretrained architectures (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024). For embedding models the common objectives are pointwise alignment to teacher coordinates, contrastive learning with teacher-derived positives, teacher-anchored layer alignment, and relational losses that match pairwise similarities within a mini-batch. Each supervises at a different granularity: individual examples, internal layers, sampled pairs, or local batch structure.

The last of these is the closest to what retrieval actually needs, and it is where this paper starts. A mini-batch is a uniform sample of the corpus, but an anchor’s semantic neighbors occupy a set of measure k/N within it, so they are almost never present. The consequence is sharper than slow learning: at the objective’s own optimum, the pairs a batch never contained are subject to no constraint whatsoever, so whatever similarity the student assigns them reflects its own inductive bias rather than the teacher’s geometry. Local geometry is the part of the teacher a batch-relational objective transfers least, and closing the gap by training longer costs a factor of order N/B in epochs.

The geometric view suggests where to look instead. A teacher embedding space can be treated as a sampled semantic manifold: nearby examples define local neighborhoods, and multi-hop paths reveal semantic continuity. If the teacher embeddings define a discrete manifold over training examples, random walks on it reveal how semantic mass flows from each anchor to nearby examples, broader regions, and higher-order communities. Matching that diffusion process gives the student geometry-aware supervision tied neither to teacher internals nor to a particular student architecture.

We propose **RIPPLE**, a manifold distillation method for compact text embedding models. RIPPLE caches teacher embeddings once, builds a mutual k -nearest-neighbor graph from teacher cosine similarity, and reads the graph as a discrete approximation of the teacher’s semantic manifold. From its transition matrix it precomputes diffusion targets across a range of diffusion times, from the ambient teacher metric itself to one-step, two-step, and longer-range semantic random walks, and trains the student to reproduce them on small candidate sets drawn in proportion to diffusion mass and pooled across the mini-batch. A second term extends supervision past the anchors: random walks select non-anchor nodes the batch has already encoded, and the student reproduces the teacher’s one-step transition row there as well, which both densifies supervision and pins down similarity offsets that anchor-only matching leaves free. After the graph is built the teacher is discarded, so training never runs a teacher forward pass.

Two design decisions keep this family from collapsing into target smoothing. Matching one-step diffusion is already enough to control every longer diffusion time, but only up to an error that grows with it, so the broader scales earn their place by removing that growth. And they only do so if the student answers each scale at its own resolution: with one shared student distribution the multi-scale objective is exactly a single-scale objective against the mixture. RIPPLE needs no pointwise regression, contrastive pair construction, layer alignment, or graph-coordinate objective, and adds no trainable parameters beyond the student encoder.

Our contributions are:

- We show that a mini-batch relational objective identifies an anchor’s similarities only on the pairs its batches happened to contain—a fraction of order TB/N of its neighbors—so at any practical schedule it supplies no teacher signal on nearly all of the local geometry it is meant to transfer (Section 3).
- We introduce RIPPLE, which distills a diffusion-time family of neighborhood distributions from a teacher-induced mutual k NN graph into a compact student encoder, and derive each of its components from the requirement it has to meet: why local matching controls global geometry, why broader scales still tighten it, why small sampled candidate sets suffice, why the scales must be matched at distinct resolutions, and why one calibration term completes the loss (Section 4).
- Across nine datasets and three teacher–student pairs, RIPPLE attains the best overall average among compact students in two of three settings and second best in the third, with the largest gains on semantic textual similarity; ablations test each theoretical prediction separately (Section 5.3).

2 RELATED WORK

Text Embedding Models. Text embedding models encode variable-length text into fixed-dimensional vectors for retrieval, clustering, and semantic textual similarity. Sentence-BERT made transformer encoders practical for sentence-level similarity search (Reimers & Gurevych, 2019), SimCSE showed that contrastive learning with dropout noise or entailment supervision produces strong sentence embeddings (Gao et al., 2021b), and evaluation suites such as BEIR and MTEB shifted the field toward broad multi-task assessment (Thakur et al., 2021; Muennighoff et al., 2023).

Recent systems scale this recipe with larger foundation models and specialized retrieval objectives: BGE-M3 combines multilingual, multi-functionality, and multi-granularity training with self-distillation across retrieval modes (Chen et al., 2024), and Qwen3 Embedding scales embedding and reranking models from Qwen3 (Zhang et al., 2025). This line supplies stronger teachers but does not specify how a compact student should inherit their geometry. RIPPLE treats such models as black-box teachers, requiring only their final embeddings.

Distillation for Language and Embedding Models. Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton et al., 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders through combinations of logits, hidden states, attention distributions, and self-attention relations (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020). Recent LLM distillation extends this toolkit with on-policy training, contrastive objectives, sparse-logit acceleration, and score-matching formulations (Agarwal et al., 2024; Ko et al., 2025; Anshumann et al., 2025; Kim et al., 2025; Xu et al., 2024), and cross-tokenizer methods such as DSKD and CDM address vocabulary mismatch (Zhang et al., 2024b; Chen et al., 2025).

Embedding-specific distillation adapts these ideas to retrieval-oriented representations. EmbedDistill aligns student and teacher retrieval geometry, including query–document embedding structure, for dense retrievers (Kim et al., 2023). Gecko distills LLM-generated and LLM-relabelled retrieval data into compact embedding models (Lee et al., 2024); Jasper and Stella combine multi-stage embedding distillation with Matryoshka Representation Learning (Zhang et al., 2024a); LEAF aligns compact students to teacher representations and supports asymmetric retrieval (Vujanic & Ruecksties, 2025); and Jina Embeddings v5 pairs task-targeted distillation with contrastive training (Akram et al., 2026). These methods establish distillation as a central recipe for embedding development. RIPPLE is complementary: rather than introducing teacher internals or task labels, it converts cached final embeddings into manifold diffusion targets.

Relational and Local Supervision. This is the family RIPPLE is positioned against. Pointwise embedding distillation matches a projected student embedding to the corresponding teacher embedding; it is architecture-agnostic but asks a compact student to reproduce absolute coordinates even when its capacity and dimension differ. Contrastive sentence-embedding distillation improves on this by transferring contrastive behavior and pairwise preferences (Gao et al., 2021a; Xu et al., 2023), and relational knowledge distillation argues that students should preserve relations among examples rather than individual outputs (Park et al., 2019). In embedding training this becomes mini-batch similarity matching: teacher and student pairwise similarity matrices are compared over the examples that happen to co-occur in a stochastic batch. Such objectives reflect retrieval better than coordinate regression, but their supervision is bounded by the sampled batch, and Section 3 shows the bound is sharper than it appears—neighbor coverage accrues at rate $O(B/N)$ per epoch, and the normalized in-batch target is biased, not merely noisy.

A second line supplies intermediate supervision when teacher internals or student layers are available, including CKA-style representation matching and optimal-transport alignment (Kornblith et al., 2019; Alqahtani et al., 2021). Closest among these is teacher-anchored layer alignment (Dao et al., 2026), our strongest baseline, which targets internal transfer under capacity mismatch. RIPPLE requires no such access: it uses only the neighborhood geometry induced by final teacher embeddings.

Graph, Manifold, and Diffusion Geometry. Graph-based manifold learning studies how neighborhood graphs reveal structure in high-dimensional data. Diffusion Maps use Markov diffusion on a graph to expose multi-scale connectivity (Coifman & Lafon, 2006), and more recently Heat-Geo relates heat diffusion to geodesic distance through Varadhan’s formula, fitting an embedding that preserves a heat-geodesic dissimilarity (Huguet et al., 2023). Knowledge distillation on graphs studies how structural information transfers between graph models or from graph teachers to simpler students (Tian et al., 2025; Zhang et al., 2022).

RIPPLE draws on this view but inverts what diffusion is used for. Manifold-learning methods fit *coordinates* for a fixed point set so that a diffusion-derived distance is preserved; RIPPLE uses diffusion rows as *targets* for a parametric encoder, so what transfers is a conditional distribution over neighbors rather than a distance to be embedded, and the result applies to text the graph never

contained. The graph is also not an observed input graph: it is induced by a strong text embedding teacher, which is what makes its diffusion distributions a supervision signal rather than an output.

3 WHAT MINI-BATCH RELATIONS LEAVE UNDETERMINED

Relational distillation compares teacher and student similarities over the examples that happen to co-occur in a stochastic mini-batch (Park et al., 2019). This section asks what such an objective determines and what it leaves free. The answer is stark: driving it to zero constrains only the pairs a batch happened to contain, an anchor’s own neighbors are almost never among them, and the rest of its local geometry may take arbitrary values without changing the loss. Both are properties of the sampling scheme rather than of the loss or the optimizer, and buying them back by training longer costs a factor of order N/B in epochs. Proofs are in Appendix A.

Setup and notation. Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be an unlabeled distillation corpus with cached teacher embeddings t_i on the unit sphere, write $c_{ij}^T = \cos(t_i, t_j)$, and let $\mathcal{N}_k(i)$ denote the k nearest neighbors of x_i under c^T . A student encoder S_θ produces embeddings s_i with scores $c_{ij}^S = \cos(s_i, s_j)$. For distributions p, q on a common finite set, $\text{TV}(p, q) = \frac{1}{2} \|p - q\|_1$, and for a subset C of the support, $p|_C$ denotes restriction to C followed by renormalization. We model an epoch as a uniformly random partition of $\mathcal{D}$ into batches of size B , resampled independently across epochs. The batch-relational objective compares, at a temperature τ_0 , the restricted distributions

$$p_i^{T,B}(j) \propto \exp(c_{ij}^T/\tau_0), \quad p_i^{S,B}(j) \propto \exp(c_{ij}^S/\tau_0), \quad j \in \mathcal{B} \setminus \{i\}, \quad (1)$$

through $\mathcal{L}_B = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} D_{\text{KL}}(p_i^{T,B} \| p_i^{S,B})$.

Proposition 1 (Neighborhood coverage). *Fix an anchor i and let $\mathcal{B} \ni i$ be its batch. Then (i) $\mathbb{E} |\mathcal{B} \cap \mathcal{N}_k(i)| = (B-1)k/(N-1)$; (ii) $\Pr[\mathcal{B} \cap \mathcal{N}_k(i) = \emptyset] \geq 1 - (B-1)k/(N-1)$; (iii) across T independently resampled epochs, the expected fraction of pairs $\{(i, j) : j \in \mathcal{N}_k(i)\}$ that ever co-occur is at most $T(B-1)/(N-1)$.*

Coverage alone would be a sample-efficiency complaint, repairable by training longer. The next result shows it is not: what a batch never contains is not merely under-supervised but entirely unconstrained, so the objective’s own optimum leaves it free.

Proposition 2 (Under-determination of batch relations). *Fix the batches realized over T epochs and let $\mathcal{S}_i^T$ be the union of those containing i , with i removed. (i) $\mathcal{L}_B = 0$ on every realized batch if and only if for each such $\mathcal{B}$ and each $i \in \mathcal{B}$ there is a constant a_i^B with $c_{ij}^S = c_{ij}^T + a_i^B$ for all $j \in \mathcal{B} \setminus \{i\}$. (ii) Consequently the scores $\{c_{ij}^S : j \notin \mathcal{S}_i^T\}$ are unconstrained: they may be set arbitrarily while the objective stays at zero. Offsets attached to two batches are independent whenever those batches share no element.*

Part (i) also shows why the defect is not a bias of the teacher target: the perfect student $c_{ij}^S = c_{ij}^T + a_i$ attains zero loss on *every* batch, because target and student are restricted to the same columns and the two restrictions cancel. What fails is identifiability. By Proposition 1(iii) the constrained fraction of an anchor’s neighbors is $T(B-1)/(N-1) \approx TB/N$: it is governed by how many examples an anchor is ever compared against over the whole run, relative to the corpus, and—usefully—it does not depend on k , so it is not an artifact of how one chooses to delimit a neighborhood. For the corpus and schedule of our experiments ($N \approx 15\text{K}$, $B = 32$, five epochs) it is near 1%, and it reaches 1 only at T of order N/B , roughly 480 epochs here against the five we train, and more once repeated draws are accounted for.

On the pairs outside that fraction the objective supplies no signal at all. This does not mean a trained student assigns them arbitrary similarities—an encoder shares parameters across all pairs, so its predictions there are perfectly well determined—but they are determined by the student’s own inductive bias rather than by the teacher. Batch-relational distillation therefore delegates local geometry to the student’s prior instead of transferring it, which is the gap Section 4 closes. Part (ii) adds that even the constrained scores carry a separate offset per batch, so similarity levels are not comparable across batches either.

The remedy is to stop letting the batch decide which comparisons an anchor sees. Section 4 builds each anchor’s neighborhood explicitly from cached teacher embeddings and supervises it at every

step, and the rest of the method follows from what that shift requires. The identifiability question returns there in a milder form—anchored matching leaves one offset per anchor rather than one per unseen neighbor—and Section 4.5 closes it.

4 RIPPLE

RIPPLE treats the teacher embedding space as a sampled semantic manifold and transfers its geometry through diffusion on that manifold. Section 4.1 builds the discrete manifold, Section 4.2 defines what the student must match, Section 4.3 the columns it is matched on, Section 4.4 the objective, and Section 4.5 what that objective leaves undetermined. Proofs are in Appendix A.

4.1 THE TEACHER MANIFOLD GRAPH

Let $t_i = T(x_i)$ be the cached embedding of x_i under teacher encoder T . Writing $\mathcal{N}_k^T(i)$ for the top- k neighbors of x_i under c^T , RIPPLE keeps only mutual edges, $j \in \mathcal{M}_k^T(i) \Leftrightarrow j \in \mathcal{N}_k^T(i)$ and $i \in \mathcal{N}_k^T(j)$, which suppresses one-directional edges and hubness. The weighted adjacency is $W_{ij}^T = \exp(\cos(t_i, t_j)/\tau_i)$ for $j \in \mathcal{M}_k^T(i)$ and 0 otherwise, where τ_i is set by entropic affinity so that every row carries the same effective number of neighbors ρ_g (Hinton & Roweis, 2002; van der Maaten & Hinton, 2008; Vladymyrov & Carreira-Perpiñán, 2013). With diagonal degree matrix D^T , the transition matrix $P^T = (D^T)^{-1}W^T$ has rows P_i^T giving the one-step semantic random walk from x_i , and $\tilde{P}^T = \frac{1}{2}(I + P^T)$ is the corresponding lazy walk.

4.2 DIFFUSION TARGETS

We read the graph as a discrete manifold in the sense of diffusion maps (Coifman & Lafon, 2006): under suitable sampling conditions, neighborhood-graph diffusion converges to the heat semigroup of the underlying manifold (von Luxburg et al., 2008; Trillos & Slepčev, 2018; Calder & Trillos, 2022). We use this as interpretation only; all statements below concern the discrete objects. Diffusion time is then the natural scale parameter, interpolating between the ambient metric and global connectivity, and RIPPLE distills a discrete analogue of that interpolation rather than a single point on it. Powers of the lazy walk give the graph scales, $P_r^T = (\tilde{P}^T)^r$ for $r \in \{1, 2, 4\}$: small r preserves sharp local neighborhoods, larger r captures multi-hop connectivity and community structure. The family’s other endpoint is the ambient distribution, the teacher metric before any propagation,

$$p_{i,0}^T(j) = \frac{\exp(c_{ij}^T/\tau_0)}{\sum_{j' \neq i} \exp(c_{ij'}^T/\tau_0)}, \quad (2)$$

and we write $\mathcal{R} = \{0, 1, 2, 4\}$ for the full family. Diffusion is computed sparsely and offline, propagating mass through neighbor lists with per-step truncation at tolerance δ , which perturbs an r -step target by at most $r\delta$ in total variation (Appendix B).

The student has no transition matrix of its own; it produces scores. We therefore read one off the graph, $P_i^S := \text{softmax}_{\mathcal{M}_k^T(i)}(c_i^S/\tau_i)$, the student’s row at the graph’s own support and bandwidth. This is the object the sharpest term of Section 4.4 controls, and with it the sharp end of the family is enough to control the rest.

Theorem 1 (Local-to-global transfer). *Let P^T and P^S be teacher and student transition matrices on the same support. Suppose for every anchor i : the restricted divergence satisfies $D_{\text{KL}}(p_{i,1}^T|_{C_i} \| p_{i,1}^S) \leq \varepsilon$ on a column set C_i (constructed in Section 4.3), the set carries teacher mass $p_{i,1}^T(C_i) \geq 1 - \delta_T$, and the student’s full-row mass outside C_i is at most δ_S . Then for every diffusion time $r \geq 1$,*

$$\max_i \text{TV}(\delta_i^\top (\tilde{P}^T)^r, \delta_i^\top (\tilde{P}^S)^r) \leq \frac{r}{2} \left(\sqrt{\varepsilon/2} + \delta_T + \delta_S \right). \quad (3)$$

Matching one-step neighborhoods within ε therefore pins down every longer-range diffusion distribution—and with it diffusion distances and community structure—without supervising them, which is what licenses purely local supervision. The bound also degrades linearly in r , and supervising a broad scale directly removes that growth.

Corollary 1 (Multi-scale tightening). *Under the conditions of Theorem 1 applied at scale r directly (restricted divergence ε_r , mass defects δ_T, δ_S at scale r), the r -step error is at most $\sqrt{\varepsilon_r/2} + \delta_T + \delta_S$, with no factor of r .*

The family is thus a trade of supervision for tightness rather than redundancy. Its ambient member, $r = 0$, is the second endpoint of the interpolation: the graph scales describe how mass *moves* on the manifold, the ambient scale the metric it moves on. Being defined for every pair rather than only on the graph support, it is also the one term that can be normalized over a column set shared by the whole batch.

4.3 SAMPLED COLUMNS

The support of $p_{i,r}^T$ grows with k and with diffusion time, so scoring every anchor against its full multi-hop neighborhood is not viable. It is also not necessary.

Proposition 3 (Sampling error). *Let p be a diffusion target for one anchor, and let the candidate set C be formed by m independent draws from p . Then*

$$\mathbb{E}[\text{TV}(p|_C, p)] = \mathbb{E}[p(\bar{C})] \leq \min_{t>0} \left(e^{-tm} + \text{Tail}_p(t) \right), \quad \text{Tail}_p(t) = \sum_{j: p_j < t} p_j. \quad (4)$$

In particular, if every atom in the support of p carries mass at least c/ρ with $\rho = e^{H(p)}$ its perplexity, the expected error is at most $e^{-cm/\rho}$.

The sample size that suffices is governed by the *sharpness* of the target rather than by the neighborhood size k , so the candidate set stays small as the graph grows. Each anchor is scored against $C_i = C_i^+ \cup C_i^h \cup C_i^-$: high-probability diffusion neighbors drawn by Gumbel top- k (Efraimidis & Spirakis, 2006; Kool et al., 2019), teacher hard negatives mined from same-source nearest neighbors outside the graph, and random negatives, all redrawn every epoch from precomputed pools. The teacher target at scale $r \geq 1$ is the diffusion row restricted and renormalized, $p_{i,r}^T = (P_r^T)_i|_{C_i}$, which expresses graded semantic relatedness rather than a binary positive–negative label. The negatives control δ_S in Theorem 1: a small restricted divergence bounds the student’s mass on the negative block, and that block is resampled from the complement of the diffusion support at every epoch (Appendix A.4).

Within a mini-batch B the candidate sets are pooled into $\mathcal{P}_B = \bigcup_{i \in B} C_i$, deduplicated and with the anchor’s own column removed—so the $r = 1$ target is exactly P_i^T restricted to C_i , the lazy walk’s self-mass being dropped with that column. Since $\mathcal{P}_B$ is encoded once for the whole batch, pooling multiplies the comparisons per anchor by roughly $|B|$ at unchanged encoding cost, and the shared columns make similarity levels comparable across the batch—the property that semantic textual similarity and a global cosine threshold both depend on. The ambient scale is normalized over $\mathcal{P}_B$, the graph scales over C_i .

4.4 THE DIFFUSION OBJECTIVE

Let $s_i = S_\theta(x_i)$. At each scale the student induces a distribution over the corresponding column set $\Omega_{i,r}$ ($\mathcal{P}_B$ for $r = 0$, C_i otherwise),

$$p_{i,r}^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_r)}{\sum_{j' \in \Omega_{i,r}} \exp(\cos(s_i, s_{j'})/\tau_r)}, \quad j \in \Omega_{i,r}, \quad (5)$$

and RIPPLE minimizes the weighted divergence to the teacher family,

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}}(p_{i,r}^T \| p_{i,r}^S). \quad (6)$$

Corollary 1 says a broad scale is worth supervising; it does not say the supervision survives being added to the sum. It does not, if the student answers every scale with the same distribution.

Proposition 4 (Collapse under a shared student distribution). *If all scales share one student distribution p_i^S , then*

$$\sum_r \omega_r D_{\text{KL}}(p_{i,r}^T \| p_i^S) = H(\bar{p}_i^T, p_i^S) - \sum_r \omega_r H(p_{i,r}^T), \quad \bar{p}_i^T = \sum_r \omega_r p_{i,r}^T, \quad (7)$$

where the second term does not depend on the student parameters.

Cross-entropy is linear in its target, so under a shared student distribution any two scale families with the same mixture produce identical gradients and multi-scale supervision degenerates into target smoothing. Eq. (5) therefore gives each scale its own τ_r . We set them by the spread of the diffusion itself, $\tau_r = \sqrt{r} \tau_1$, so that each scale is matched at the resolution its own target already has, and we weight them by $\omega_r = 1/r$ with $\omega_0 = \omega_1$. Matching the same similarities at several resolutions is over-determined, which forces the student to reproduce the teacher’s cosine *gaps* rather than only its ranking—the information graded similarity tasks measure. The anchor of that ladder is not a convention either: Theorem 1 is a statement about P_i^S , the student’s row at the graph’s own bandwidth, so the $r = 1$ term must be taken at $\tau_1(i) = \tau_i$ for the theorem to apply to the quantity actually optimized, and the row term of Section 4.5 inherits the same requirement (Appendix B). Fixing τ_1 fixes the ladder above it, so no student temperature on the diffusion scales is chosen; τ_0 at the ambient scale is the only one that is.

4.5 THE RESIDUAL GAUGE, AND ROW SUPERVISION

Every term of Eq. (6) is a softmax divergence, and a softmax is invariant to a constant shift of its logits. This leaves the objective blind to one scalar per supervised row.

Proposition 5 (Gauge of the diffusion loss). *$\mathcal{L}_{\text{diff}}$ is invariant under $c_{ij}^S \mapsto c_{ij}^S + a_i$ for arbitrary per-anchor constants a_i . Moreover, if the ambient-scale term vanishes for anchor i on a column set with at least two elements, then $c_{ij}^S = c_{ij}^T + a_i$ on that set for some $a_i \in \mathbb{R}$.*

Distribution matching thus determines all similarity *gaps* but leaves one origin free per anchor, and downstream uses are not invariant to it: a retrieval threshold or a similarity calibration is shared across anchors, so offsets $\{a_i\}$ reorder cross-anchor comparisons while leaving $\mathcal{L}_{\text{diff}}$ untouched (Remark 7). The freedom is one constant per supervised row, so it shrinks when more rows are supervised and those rows overlap—and cosine similarity is symmetric, so two rows that each appear in the other’s column set cannot carry different offsets.

Rows at non-anchor nodes are therefore what the objective is missing, and most of them are already encoded: on every step $\mathcal{P}_B$ contains several hundred non-anchor texts used only as columns. Section 4.3 sampled *columns* for a given row; the question here is which *rows* to supervise at all, and the answer is the one that also builds the overlaps the previous paragraph needs. From each anchor $i \in B$ we run M non-backtracking random walks of length L on the teacher transition matrix,

$$j_0^{(m)} = i, \quad j_{t+1}^{(m)} \sim P_{j_t^{(m)}}^T, \quad t = 0, \dots, L-1, \quad (8)$$

and supervise the nodes visited at steps $t \geq 1$, weighted by their normalized visit counts ν_B . Step 0 is the anchor, whose row is already the $r = 1$ target. Visited nodes not already in C_i replace random negatives, so the pooled set covers the walk without enlarging it. For a visited node j the columns are the teacher’s own support for that row intersected with the batch, $\Omega_j^w = \mathcal{P}_B \cap \mathcal{M}_k^T(j)$, so that every column carries teacher mass; with student distributions $p_{w,j}^S(j') \propto \exp(\cos(s_j, s_{j'})/\tau_j)$ on Ω_j^w ,

$$\mathcal{L}_{\text{row}} = \sum_j \nu_B(j) D_{\text{KL}}\left(P_j^T|_{\Omega_j^w} \parallel p_{w,j}^S\right). \quad (9)$$

Like the sharpest graph scale, this term matches a transition row at its own bandwidth, so its infimum is 0 and it is on the same scale as Eq. (6).

Writing $\mathcal{S}$ for the set of supervised rows, Ω_u for the columns of row u , and $\mathcal{G}$ for the graph on $\mathcal{S}$ with an edge $\{u, v\}$ whenever $v \in \Omega_u$ and $u \in \Omega_v$:

Proposition 6 (Gauge reduction by row supervision). *Assume $\tau_w(u) = \tau_u$ and $|\Omega_u| \geq 2$ for every $u \in \mathcal{S}$, and suppose every supervised term vanishes: the ambient term of Eq. (6) at each anchor, and the row term of Eq. (9) at each visited node. Then there are constants $\{a_u\}$ with $c_{uv}^S = c_{uv}^T + a_u$ for every $v \in \Omega_u$, and $a_u = a_v$ for every edge of $\mathcal{G}$. Consequently a is constant on each connected component, and the invariance group of the objective is $\mathbb{R}^\kappa$ with κ the number of components, rather than $\mathbb{R}^{|\mathcal{S}|}$.*

The walk builds the edges of $\mathcal{G}$ as it moves: consecutive visited nodes are adjacent in the mutual- k NN graph, adjacency is symmetric, and both endpoints are supervised, so each traversed step

contributes an edge. It is the trajectory, not merely the set of visited nodes, that does this work, and how fast κ falls is a coverage question of exactly the form already solved in Section 4.3.

Proposition 7 (Walk coverage). *Let q be the distribution over edges of $\mathcal{M}_k^T$ traversed by the walks in one optimization step. After n steps the expected fraction of edges never traversed is*

$$\sum_e q_e (1 - q_e)^n \leq \min_{t \geq 0} \left(e^{-tn} + \text{Tail}_q(t) \right). \quad (10)$$

Since the components of $\mathcal{G}$ are determined by the traversed edges, κ is non-increasing in n and is at most the number of components of the traversed subgraph.

This is Proposition 3 applied to edges rather than columns: one bound governs both how fast a candidate set covers a neighborhood and how fast walks connect the constraint graph. Non-backtracking raises the number of distinct edges per unit of walk budget, hence the effective n .

The residual freedom is also benign in a way the per-anchor freedom is not: one offset shared by a component shifts all of its cosines equally, preserving every cross-anchor ordering and displacing only the absolute location of a shared threshold.

Remark 1 (Approximate identification, and what bounds L). Proposition 6 assumes exact zeros. If instead every supervised term is at most ε and the teacher’s restricted rows carry mass at least $p_{\min}$ on each supervised column, then two offsets joined by a traversed edge differ by $O(\tau\sqrt{\varepsilon}/p_{\min})$, and the discrepancy accumulates along a path of ℓ edges (Appendix A.8). Longer walks therefore cover more edges by Proposition 7 but propagate offset error further, so L trades coverage against accumulation and its optimum is interior.

4.6 FULL OBJECTIVE

$$\mathcal{L}_{\text{RIPPLE}} = \mathcal{L}_{\text{diff}} + \lambda \mathcal{L}_{\text{row}} \quad (11)$$

Training requires only cached teacher embeddings—no teacher hidden states, no online teacher pass, no task labels, no contrastive or layer-alignment term, and no change to the student architecture. After the graph is built the teacher is discarded.

5 EXPERIMENTS

5.1 SETUP

Models. We evaluate three teacher $\rightarrow$ student pairs: Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M), BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M), and Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M). Teacher embeddings are cached once before training; after graph construction the teacher is removed from GPU memory, and student training uses only precomputed diffusion targets and the corresponding candidate texts.

Training data. Following the TALAS setup (Dao et al., 2026), we train on approximately 15K unlabeled sentences sampled evenly from the three in-domain datasets Emotion, WiC, and STS-B. RIPPLE constructs the teacher semantic graph directly over these texts.

Evaluation. We evaluate on three task families: text classification with F1 on Banking77, Tweet, and Emotion; pair classification with average precision (AP) on MRPC, SciTail, and WiC; and semantic textual similarity with Spearman correlation on SICK, STS12, and STS-B. Emotion, WiC, and STS-B are in-domain, the other six out-of-domain. Avg-In and Avg-Out are the unweighted means over their respective three and six datasets, and Avg. is the unweighted mean over all nine.

Baselines. We compare against the teacher, the undistilled student, SimCSE-unsup, CDM, DSKD, Jasper and Stella, DistillCSE, EMO, and TALAS, using the numbers reported by TALAS (Dao et al., 2026) for the same three configurations. To test Section 3 directly we additionally train a controlled *batch-relational* arm: identical corpus, student, schedule, and code path, with the objective replaced by mini-batch teacher–student similarity matching at τ_0 . This isolates the anchored-versus-batch comparison from every other difference between methods.

Implementation. The teacher graph uses $k = 200$ mutual neighbors, and diffusion is taken over $\mathcal{R} = \{0, 1, 2, 4\}$ with weights $\omega_r = 1/r$ ($\omega_0 = \omega_1$), normalized. Each anchor draws a candidate set of size 64 from diffusion neighbors, teacher hard negatives, and random negatives, redrawn every epoch and pooled across mini-batches of 32 anchors. The only student temperature we choose is $\tau_0 = 0.10$ at the ambient scale; the sharpest scale and the row term take the bandwidth of the row they match, and the broader scales follow $\tau_r = \sqrt{r} \tau_1$ (Appendix B). Row supervision draws $M = 4$ non-backtracking walks of length $L = 4$ per anchor per epoch at $\lambda = 0.5$, enabled from the second epoch onward. The truncation tolerance is $\delta = 10^{-2}$. Training runs for five epochs at a learning rate of 3×10^{-5} ; remaining settings are in the released configuration.

The graph in the runs below is built with a single bandwidth $\tau_g = 0.05$ shared by every row rather than the entropic affinity of Section 4.1; entropic-bandwidth runs at $\rho_g = 30$ are in progress and will be reported as the corresponding row of Table 2.

5.2 MAIN RESULTS

Table 1 reports all three teacher–student settings across the nine datasets. RIPPLE attains the best overall average among compact students in two of the three settings, 77.98 against 76.55 for the strongest baseline when distilling BGE-M3 into MiniLMv2-H768, and 75.33 against 74.79 when distilling Qwen3-Embedding-0.6B into the 22M MiniLMv2-H384. In the Qwen3-Embedding-4B to BERT-base setting it reaches 78.56, ranking second behind TALAS at 78.83.

The advantage is concentrated where the objective operates. RIPPLE gives the best student result on Banking77 and on STS-B in all three settings, improving STS-B by 2.26, 2.75, and 0.68 Spearman points over the strongest baseline, and it leads the STS family on all three datasets in both MiniLMv2 settings. It also obtains the best in-domain average in the two settings it wins overall, by 2.81 and 1.91 points, with the largest single-dataset gains on Emotion (+6.20 and +6.19 F1). The margins grow as the student shrinks: matching diffusion distributions rather than individual teacher coordinates asks the student to preserve relative neighborhood structure, which remains attainable when embedding capacity is well below the teacher’s. WiC is the one dataset where RIPPLE trails consistently, reflecting that word-in-context judgments depend on token-level sense disambiguation rather than the sentence-level neighborhood geometry the diffusion targets describe.

5.3 ABLATIONS

Table 2 removes one design decision at a time, each mapped to the result that motivates it; Table 3 sweeps the quantities the bounds predict should saturate. All runs use the BGE-M3 $\rightarrow$ MiniLMv2-H768 setting and the nine-dataset protocol of Table 1.

Anchored versus batch supervision. Propositions 1 and 2 predict that a batch-relational objective supplies no teacher signal on nearly all of each anchor’s local geometry, while determining coarse structure normally. The controlled arm shares everything but the objective, so the gap isolates that prediction: the loss should concentrate on tasks that depend on fine neighborhood structure rather than on class separation. We also report δ_S , the student’s mass outside the candidate set, which Theorem 1 assumes small and which is directly measurable on a trained model.

The scale family and its resolutions. Theorem 1 says $r = 1$ alone already controls every diffusion time, so a single-scale arm should be a strong ablation rather than a broken one; Corollary 1 predicts the broad scales still help, by removing the factor r from the bound at the ranges retrieval browses. Proposition 4 makes a sharper prediction: with a shared student temperature the multi-scale objective is exactly a single-scale one against the mixture, so the *shared- τ* arm should recover little of the gain that the full family provides. Removing the ambient scale ($\omega_0 = 0$) tests the three roles claimed for it in Section 4.2, and by Proposition 5 should cost most where cross-anchor calibration matters.

Row supervision and the similarity gauge. Propositions 5 and 6 make a falsifiable prediction. Without $\mathcal{L}_{\text{row}}$ the supervised rows are the anchors alone, the constraint graph has no edges, and one similarity offset per anchor is free; with it, each traversed edge identifies two offsets and the freedom collapses to one constant per connected component. Tasks that compare cosines *across* anchors against a shared scale—pair classification with a global threshold, and cross-anchor rank correlation

Table 1: Results across three teacher $\rightarrow$ student settings on nine datasets. Emotion, WiC, and STS-B (marked with $\star$) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best compact-model results within each setting. Baseline results are from TALAS (Dao et al., 2026). For setting (c), we report the epoch checkpoint with the highest overall average.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg.
	Banking77	Tweet	Emotion $\star$	MRPC	SciTail	WiC $\star$	SICK	STS12	STS-B $\star$	Avg-In	Avg-Out	Avg.
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	<u>71.53</u>	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	<u>73.83</u>	65.98	<u>85.33</u>	80.99	<u>70.50</u>	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	<u>91.43</u>	74.30	70.71	86.47	82.66	69.24	78.38	75.40	80.88	73.61	81.44	78.83
RIPPLE	91.86	73.06	<u>69.20</u>	84.72	86.19	67.19	<u>77.54</u>	<u>74.19</u>	83.14	<u>73.17</u>	<u>81.26</u>	<u>78.56</u>
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	44.24	62.41	56.35
SimCSE-unsup	87.89	71.45	55.65	83.87	76.32	67.96	66.93	59.66	63.97	62.53	74.35	70.41
CDM	89.15	70.48	58.71	84.16	75.91	70.00	68.14	62.74	67.78	65.50	75.10	71.90
DSKD	89.51	70.53	57.27	84.39	75.36	<u>69.96</u>	68.81	66.78	70.05	65.76	75.90	72.52
Jasper and Stella	88.38	74.13	62.81	<u>85.98</u>	80.33	68.38	74.35	66.80	74.55	68.58	78.33	75.08
DistillCSE	90.50	72.70	59.88	84.39	78.48	69.49	73.97	<u>70.00</u>	73.73	67.70	78.34	74.79
EMO	90.19	<u>74.26</u>	61.48	85.31	80.82	68.30	75.41	67.06	76.35	68.71	78.84	75.46
TALAS	<u>90.62</u>	74.81	<u>63.30</u>	87.05	81.94	66.28	<u>76.86</u>	69.04	<u>79.01</u>	<u>69.53</u>	<u>80.05</u>	<u>76.55</u>
RIPPLE	90.86	73.08	69.50	85.74	83.57	65.77	77.32	74.18	81.76	72.34	80.79	77.98
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	40.91	58.45	52.60
SimCSE-unsup	86.67	69.56	54.00	82.29	73.76	65.71	66.04	59.11	62.08	60.60	72.91	68.80
CDM	86.07	70.32	53.04	81.97	72.97	68.34	65.97	61.12	65.75	62.38	73.07	69.51
DSKD	85.66	69.86	52.25	82.17	73.52	<u>68.58</u>	66.54	62.52	66.70	62.51	73.38	69.76
Jasper and Stella	85.94	71.25	57.99	83.36	76.17	68.25	70.57	63.81	70.11	65.45	75.18	71.94
DistillCSE	<u>88.03</u>	69.87	54.24	83.20	77.66	67.88	70.09	68.19	71.53	64.55	76.17	72.30
EMO	87.03	<u>71.67</u>	59.63	<u>83.45</u>	78.17	68.89	70.81	65.78	71.42	66.65	76.15	72.98
TALAS	86.69	72.77	<u>60.94</u>	85.10	81.42	66.65	<u>72.51</u>	<u>70.77</u>	<u>76.22</u>	<u>67.94</u>	78.21	<u>74.79</u>
RIPPLE	88.31	70.41	67.13	83.00	<u>80.60</u>	65.51	73.98	72.14	76.90	69.85	<u>78.07</u>	75.33

in STS—should therefore degrade when the term is removed, while within-anchor structure such as classification by nearest neighborhoods should be comparatively unaffected. Table 2 additionally reports the dispersion of the estimated per-anchor offsets $\hat{a}_i = \sum_j p_{i,0}^T(j) [\cos(s_i, s_j) - \cos(t_i, t_j)]$, which the theory predicts to widen without row supervision, together with κ , the number of components of the realized constraint graph.

Saturation. Proposition 3 bounds the error of scoring each anchor on a sampled candidate set by the uncovered target mass, which decays with m at a rate set by the sharpness of the diffusion target rather than by k , so performance should saturate at a candidate size well below the neighborhood size. Theorem 1 transfers geometry from whichever graph the teacher induces, and the manifold interpretation behind it improves as neighborhoods grow—until they cut across semantic boundaries—so k should rise and then plateau or decline rather than improve monotonically. For the walk the two results pull against each other: Proposition 7 favors long walks, which traverse more edges and merge more components of $\mathcal{G}$, while Remark 1 penalizes them, since an offset identified through a chain of ℓ edges accumulates ℓ times the per-edge error. The prediction for L is therefore an interior optimum rather than a plateau, and κ should fall monotonically with L even where the nine-dataset average does not. Since Eq. (9) is a divergence on the same scale as Eq. (6), λ is a genuine trade-off rather than a units conversion.

Table 2: Objective ablations (BGE-M3 $\rightarrow$ MiniLMv2-H768). Offset dispersion is the standard deviation of $\hat{a}_i$ across anchors; κ is the number of connected components of the realized constraint graph. Each row names the result it tests.

	Tests	Class. (F1)	Pair (AP)	STS (Spear.)	Avg.	Offset disp.	κ
RIPPLE (full)	—	[TBD]	[TBD]	[TBD]	77.98	[TBD]	[TBD]
Batch-relational arm	Prop. 2	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
single scale $\mathcal{R} = \{1\}$	Thm. 1	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
sharp only $\mathcal{R} = \{0, 1\}$	Cor. 1	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
shared student τ	Prop. 4	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
no ambient scale ($\omega_0 = 0$)	Sec. 4.2	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
w/o $\mathcal{L}_{\text{row}}$	Prop. 6	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
fixed bandwidth τ_g	Sec. 4.1	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Table 3: Sweeps of the quantities the bounds govern (BGE-M3 $\rightarrow$ MiniLMv2-H768), reporting the nine-dataset average. Defaults in bold.

Candidate size m	16	32	64	128	256
Avg.	[TBD]	[TBD]	77.98	[TBD]	[TBD]
Graph size k	50	100	200	400	
Avg.	[TBD]	[TBD]	77.98	[TBD]	
Walk length L	2	4	6	8	
Avg.	[TBD]	77.98	[TBD]	[TBD]	
Walk weight λ	0.1	0.3	0.5	1.0	
Avg.	[TBD]	[TBD]	77.98	[TBD]	

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A PROOFS

Throughout, $\|v\|_1 = \sum_j |v_j|$ and $\text{TV}(p, q) = \frac{1}{2} \|p - q\|_1$. For a nonnegative vector v with $\|v\|_1 > 0$ and a subset C of its support, $v|_C$ denotes restriction to C followed by renormalization.

A.1 PROOF OF PROPOSITION 1 (NEIGHBORHOOD COVERAGE)

Within an epoch the corpus is partitioned uniformly at random into batches of size B . By exchangeability, the batch containing i contains $B - 1$ of the remaining $N - 1$ examples, uniformly without replacement, so for any fixed $j \neq i$,

$$\Pr[j \in \mathcal{B}] = \frac{B - 1}{N - 1}. \quad (12)$$

(i) follows by linearity over the k neighbors. (ii) is Markov’s inequality applied to the nonnegative count $|\mathcal{B} \cap \mathcal{N}_k(i)|$: $\Pr[\text{count} \geq 1] \leq \mathbb{E}[\text{count}]$. For (iii), epochs are independent, so with $p_e = (B - 1)/(N - 1)$ the probability that a given neighbor pair co-occurs within T epochs is $1 - (1 - p_e)^T \leq Tp_e$, and the expected fraction of the k neighbor pairs of anchor i observed within T epochs is the same bound by linearity. $\square$

A.2 PROOF OF PROPOSITION 2 (UNDER-DETERMINATION)

(i) Fix a realized batch $\mathcal{B}$ and $i \in \mathcal{B}$. The divergence in $\mathcal{L}_{\mathcal{B}}$ vanishes exactly when the two distributions of Eq. (1) coincide on $\mathcal{B} \setminus \{i\}$. Taking log-ratios of any two columns j, j' of that set,

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_0} = \log \frac{p_i^{T, \mathcal{B}}(j)}{p_i^{T, \mathcal{B}}(j')} = \frac{c_{ij}^T - c_{ij'}^T}{\tau_0}, \quad (13)$$

so $c_{ij}^S - c_{ij'}^S$ is constant in j on $\mathcal{B} \setminus \{i\}$; call it $a_i^{\mathcal{B}}$. Conversely, if $c_{ij}^S = c_{ij}^T + a_i^{\mathcal{B}}$ on that set then the factor $e^{a_i^{\mathcal{B}}/\tau_0}$ cancels in the normalization and the two distributions coincide, so the divergence is zero.

(ii) The characterization in (i) involves c_{ij}^S only for $j \in \mathcal{B} \setminus \{i\}$, and $\mathcal{S}_i^T$ is the union of these sets over the batches containing i . Scores to columns outside $\mathcal{S}_i^T$ appear in no constraint and may be set arbitrarily. If two batches $\mathcal{B}, \mathcal{B}'$ containing i share no element, no column is subject to both constraints, so $a_i^{\mathcal{B}}$ and $a_i^{\mathcal{B}'}$ are unrelated; if they share a column j , then both equal $c_{ij}^S - c_{ij}^T$ and coincide. $\square$

Remark 2 (The defect is identifiability, not bias). The in-batch teacher target $p_i^{T, \mathcal{B}}$ is a poor estimator of the teacher’s ambient distribution over the corpus, but this does not by itself impugn the objective: the student is restricted to the same columns, so by part (i) the exact solution $c^S = c^T + a_i$ drives the loss to zero on every batch and the two restrictions cancel. The failure is that almost nothing is constrained, which Proposition 1(iii) quantifies.

A.3 TWO LEMMAS BEHIND THEOREM 1

The theorem combines a perturbation bound, which propagates a one-step error to r steps, with a restriction bound, which converts control on a column set into control of the full row.

Lemma 1 (One step to r steps). *Let P, Q be row-stochastic matrices with lazy walks $\tilde{P}, \tilde{Q}$, and suppose $\max_i \text{TV}(P_i, Q_i) \leq \varepsilon$. Then for every $r \geq 1$, $\max_i \text{TV}(\delta_i^\top \tilde{P}^r, \delta_i^\top \tilde{Q}^r) \leq r\varepsilon/2$.*

Proof. Write $\tilde{P} - \tilde{Q} = \frac{1}{2}(P - Q)$ and telescope:

$$\tilde{P}^r - \tilde{Q}^r = \sum_{s=0}^{r-1} \tilde{P}^s (\tilde{P} - \tilde{Q}) \tilde{Q}^{r-1-s}. \quad (14)$$

Fix a row i and a summand s . The vector $\mu = \delta_i^\top \tilde{P}^s$ is a probability vector, so $\mu(\tilde{P} - \tilde{Q})$ is a convex combination of the rows of $\tilde{P} - \tilde{Q}$ and

$$\|\mu(\tilde{P} - \tilde{Q})\|_1 \leq \max_j \|\frac{1}{2}(P_j - Q_j)\|_1 = \max_j \text{TV}(P_j, Q_j) \leq \varepsilon. \quad (15)$$

Right-multiplication by a row-stochastic matrix does not increase the ℓ_1 norm: $\|\nu \tilde{Q}^{r-1-s}\|_1 \leq \|\nu\|_1$. Summing the r terms gives $\|\delta_i^\top (\tilde{P}^r - \tilde{Q}^r)\|_1 \leq r\varepsilon$, i.e. $\text{TV} \leq r\varepsilon/2$. $\square$

Lemma 2 (Restriction). *Let p, q be distributions on a common finite set and C a subset with $p(C) \geq 1 - \delta_T$ and $q(C) \geq 1 - \delta_S$. Then $\text{TV}(p, q) \leq \text{TV}(p|_C, q|_C) + \delta_T + \delta_S$.*

Proof. Split the ℓ_1 distance over C and $\bar{C}$. Outside C , $\sum_{\bar{C}} |p_j - q_j| \leq p(\bar{C}) + q(\bar{C}) \leq \delta_T + \delta_S$. Inside C , write $p_j = p(C) p|_C(j)$ and $q_j = q(C) q|_C(j)$:

$$\sum_C |p_j - q_j| \leq p(C) \sum_C |p|_C(j) - q|_C(j)| + |p(C) - q(C)| \leq 2 \text{TV}(p|_C, q|_C) + \delta_T + \delta_S. \quad (16)$$

Adding the two parts and halving yields the claim. $\square$

A.4 LEAKAGE CONTROL: WHY NEGATIVES BOUND δ_S

The teacher-side defect δ_T in Theorem 1 is a property of the candidate construction; the student-side defect δ_S is what the negatives of Section 4.3 target. Let $\tilde{p} = p|_C$ and $\tilde{q} = q|_C$ be the restricted

teacher and student distributions on $C = C^+ \cup C^-$, and suppose $D_{\text{KL}}(\tilde{p} \parallel \tilde{q}) \leq \varepsilon$ with $\tilde{p}(C^-) \leq \varepsilon_{\text{neg}}$. The partition induces binary distributions $(\tilde{p}(C^+), \tilde{p}(C^-))$ and $(\tilde{q}(C^+), \tilde{q}(C^-))$, and by the data-processing inequality

$$D_{\text{KL}}(\tilde{p} \parallel \tilde{q}) \geq d_{\text{KL}}(\tilde{p}(C^+) \parallel \tilde{q}(C^+)), \quad (17)$$

where d_{KL} is the binary divergence. Pinsker's inequality for the binary pair gives $|\tilde{p}(C^+) - \tilde{q}(C^+)| \leq \sqrt{\varepsilon/2}$, hence $\tilde{q}(C^-) \leq \tilde{p}(C^-) + \sqrt{\varepsilon/2} \leq \varepsilon_{\text{neg}} + \sqrt{\varepsilon/2}$. $\square$

Remark 3 (From sampled negatives to the full complement). The bound above controls the student's mass on the negative columns actually drawn. The random negatives are uniform draws from the complement of the diffusion support, redrawn every epoch, so over training the student's scores are constrained on a fresh uniform sample of the complement at every epoch: by exchangeability, a student that hides a constant fraction of softmax mass on some subset of the complement would place mass on the drawn negatives in expectation proportional to that fraction, contradicting a uniformly small restricted loss. A quantitative uniform-convergence version of this argument (a bound on δ_S in expectation over the negative-sampling process) is deferred to the final version. Theorem 1 accordingly takes δ_S as a hypothesis.

A.5 PROOF OF THEOREM 1 (LOCAL-TO-GLOBAL TRANSFER)

Fix an anchor i . Pinsker's inequality applied to the restricted divergence gives $\text{TV}(p_{i,1}^T |_{C_i}, p_{i,1}^S) \leq \sqrt{\varepsilon/2}$. Lemma 2 with $p = p_{i,1}^T$, q the student's full row, and $C = C_i$ then bounds the full one-step rows:

$$\text{TV}(P_i^T, P_i^S) \leq \sqrt{\varepsilon/2} + \delta_T + \delta_S =: \varepsilon_{\text{row}}. \quad (18)$$

Applying Lemma 1 with row-wise bound ε_{row} yields, for every $r \geq 1$, $\max_i \text{TV}(\delta_i^\top (\tilde{P}^T)^r, \delta_i^\top (\tilde{P}^S)^r) \leq r \varepsilon_{\text{row}}/2$. $\square$

Corollary 1 is the same argument applied directly to the scale- r family: Pinsker plus Lemma 2 bound the r -step rows without invoking Lemma 1, so no factor of r appears.

Remark 4 (Average-case variant). Training controls the divergence averaged over anchors rather than its maximum. By Jensen's inequality, $\frac{1}{N} \sum_i \text{TV}(p_{i,1}^T |_{C_i}, p_{i,1}^S) \leq \sqrt{\bar{\varepsilon}/2}$ with $\bar{\varepsilon}$ the average restricted divergence, and the telescoping argument of Lemma 1 can be carried out under any fixed initial distribution over anchors, giving the same bounds for anchor-averaged errors.

A.6 PROOF OF PROPOSITION 4 (COLLAPSE)

With one shared student distribution p_i^S ,

$$\sum_r \omega_r D_{\text{KL}}(p_{i,r}^T \parallel p_i^S) = \sum_r \omega_r [H(p_{i,r}^T, p_i^S) - H(p_{i,r}^T)] = H\left(\sum_r \omega_r p_{i,r}^T, p_i^S\right) - \sum_r \omega_r H(p_{i,r}^T), \quad (19)$$

using that cross-entropy $H(p, q) = -\sum_j p_j \log q_j$ is linear in its first argument and $\sum_r \omega_r = 1$. The subtracted entropy term does not depend on the student. $\square$

A.7 TRUNCATED WALKS AND PROOF OF PROPOSITION 3

The diffusion targets are computed by a truncated walk; its error is additive across steps.

Lemma 3 (Truncation error). *If the walk is computed with per-step truncation to the highest-mass columns followed by renormalization, dropping mass fraction δ_s at step s , then after r steps the computed distribution $\hat{p}$ satisfies $\text{TV}(\hat{p}, p) \leq \sum_{s \leq r} \delta_s$.*

Proof. First, for any nonnegative vector v with total mass 1 and truncation $R(v)$ that zeroes a set of mass δ and renormalizes,

$$\text{TV}(R(v), v) = \frac{1}{2} \left[\sum_{\text{kept}} \left(\frac{v_j}{1-\delta} - v_j \right) + \sum_{\text{dropped}} v_j \right] = \frac{1}{2} [\delta + \delta] = \delta. \quad (20)$$

Let v_s denote the exact walk after s steps and $\hat{v}_s$ the truncated walk, with $M = \tilde{P}$ the (row-stochastic) step operator and δ_s the mass fraction dropped at step s . Then

$$\text{TV}(\hat{v}_s, v_s) \leq \text{TV}(R(\hat{v}_{s-1}M), \hat{v}_{s-1}M) + \text{TV}(\hat{v}_{s-1}M, v_{s-1}M) \leq \delta_s + \text{TV}(\hat{v}_{s-1}, v_{s-1}), \quad (21)$$

using the triangle inequality and that a stochastic matrix does not increase total variation. Induction over $s \leq r$ gives $\text{TV}(\tilde{v}_r, v_r) \leq \sum_{s \leq r} \delta_s$. $\square$

Proof of Proposition 3. The identity $\text{TV}(p|_C, p) = p(\bar{C})$ follows from the first display above with $\delta = p(\bar{C})$. For m independent draws from p ,

$$\mathbb{E}[p(\bar{C})] = \sum_j p_j (1 - p_j)^m. \quad (22)$$

Fix $t > 0$ and split the sum: atoms with $p_j \geq t$ contribute at most $(1 - t)^m \sum_j p_j \leq e^{-tm}$, and atoms with $p_j < t$ contribute at most $\text{Tail}_p(t)$. Minimizing over t gives the bound. If every atom in the support of p carries mass at least c/ρ , take $t = c/\rho$: no atom falls below t , the tail term vanishes, and the bound is $e^{-cm/\rho}$. $\square$

Proof of Proposition 7. Index the edges of $\mathcal{M}_k^T$ and let q_e be the probability that edge e is traversed in a given optimization step, independently across steps. The probability that e is never traversed in n steps is $(1 - q_e)^n$, so the expected q -mass of untraversed edges is $\sum_e q_e (1 - q_e)^n$, which is the quantity bounded in Proposition 3 with p replaced by q and m by n ; the same split at a threshold t gives the same bound. Adding an edge to the traversed set can only merge two components of $\mathcal{G}$ or leave them unchanged, so κ is non-increasing in n . $\square$

Remark 5 (Sampling without replacement). The implementation draws candidates by Gumbel top- k , which realizes weighted sampling without replacement (Efraimidis & Spirakis, 2006; Kool et al., 2019). Draws are distinct, so after m draws the covered mass is at least that of m independent draws in the natural coupling that re-draws duplicates; the bound of Proposition 3 therefore remains valid.

A.8 PROOFS OF PROPOSITIONS 5 AND 6 (GAUGE AND ITS REDUCTION)

Invariance. For any temperature τ and column set Ω , $\text{softmax}((c_i + a_i)/\tau)_{j \in \Omega} = \text{softmax}(c_i/\tau)_{j \in \Omega}$, since the constant $e^{a_i/\tau}$ cancels between numerator and normalizer. Every term of $\mathcal{L}_{\text{diff}}$ is a divergence between a fixed teacher target and such a softmax, so the loss is unchanged.

Zero set. Suppose the ambient-scale term vanishes for anchor i on the column set Ω , $|\Omega| \geq 2$: then $\text{softmax}(c_{ij}^S/\tau_0) = \text{softmax}(c_i^T/\tau_0)$ on Ω . Taking log-ratios of any two columns $j, j' \in \Omega$,

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_0} = \log \frac{p_{i,0}^T(j)}{p_{i,0}^T(j')} = \frac{c_{ij}^T - c_{ij'}^T}{\tau_0}, \quad (23)$$

so $c_{ij}^S - c_{ij'}^T$ is constant in j on Ω ; call it a_i . This proves Proposition 5.

Walk rows carry the same form of constraint. The teacher’s transition row is itself a softmax of teacher cosines at that row’s bandwidth, $P_j^T(j') \propto \exp(c_{jj'}^T/\tau_j)$ on $\mathcal{M}_k^T(j)$, and restricting a softmax to a subset and renormalizing is the softmax over that subset. Hence $P_j^T|_{\Omega_j^w} = \text{softmax}_{\Omega_j^w}(c_{j\cdot}^T/\tau_j)$, and the row term of Eq. (9) vanishes at j exactly when $\text{softmax}_{\Omega_j^w}(c_{j\cdot}^S/\tau_w(j)) = \text{softmax}_{\Omega_j^w}(c_{j\cdot}^T/\tau_j)$. With $\tau_w(j) = \tau_j$ the log-ratio argument above applies verbatim on Ω_j^w — the common temperature cancels from both sides of the ratio, so its value is immaterial and only the tie matters — giving $c_{jv}^S = c_{jv}^T + a_j$ for all $v \in \Omega_j^w$. The argument is thus unchanged whether the bandwidths are shared or solved per row.

Symmetry closes the gauge along edges. Every supervised row $u \in \mathcal{S}$ therefore satisfies $c_{uv}^S = c_{uv}^T + a_u$ on Ω_u , whether it was supervised by the ambient term or by the row term. Let $\{u, v\}$ be an edge of $\mathcal{G}$, so $v \in \Omega_u$ and $u \in \Omega_v$. Applying the identity in both directions,

$$c_{uv}^S = c_{uv}^T + a_u, \quad c_{vu}^S = c_{vu}^T + a_v. \quad (24)$$

Cosine similarity is symmetric, $c_{uv}^S = c_{vu}^S$ and $c_{uv}^T = c_{vu}^T$, so subtracting gives $a_u = a_v$. A function constant across every edge is constant on every connected component, so a takes at most κ distinct values, one per component of $\mathcal{G}$. Conversely, adding a common constant to all cosines within a component leaves every softmax in the objective unchanged, so the invariance group is exactly $\mathbb{R}^\kappa$. This proves Proposition 6. $\square$

Approximate identification (Remark 1). Suppose a supervised term at row u has divergence at most ε rather than zero, with teacher target p and student distribution q on Ω_u , both softmaxes at the same temperature τ . Pinsker gives $\|p - q\|_\infty \leq \text{TV}(p, q) \leq \sqrt{\varepsilon/2} =: \eta$. For any column j with $p_j \geq p_{\min} > \eta$,

$$|\log q_j - \log p_j| = \left| \log \left(1 + \frac{q_j - p_j}{p_j} \right) \right| \leq \frac{\eta}{p_{\min} - \eta}, \quad (25)$$

so for two columns j, j' the log-ratio identity used above holds up to $2\eta/(p_{\min} - \eta)$, giving $|(c_{uj}^S - c_{uj'}^S) - (c_{uj}^T - c_{uj'}^T)| \leq 2\tau\eta/(p_{\min} - \eta)$. Writing a_u for the offset fitted at row u , the same estimate bounds $|a_u - a_v|$ across an edge $\{u, v\}$ of $\mathcal{G}$ by $O(\tau\sqrt{\varepsilon}/p_{\min})$. Along a path of ℓ edges the bounds add, so offsets identified through a long chain are looser than those identified through a short one: coverage of $\mathcal{G}$ improves with walk length while the accuracy of each identification degrades with it.

Remark 6 (The walk lays down the edges). Consecutive nodes j_t, j_{t+1} of a walk with $t \geq 1$ are adjacent in $\mathcal{M}_k^T$, which is symmetric by construction, and both are supervised rows whenever both lie in $\mathcal{P}_B$. Then $j_{t+1} \in \Omega_{j_t}^w$ and $j_t \in \Omega_{j_{t+1}}^w$, so the step is an edge of $\mathcal{G}$: a walk of length L contributes a path of $L - 1$ edges. Without the walk term $\mathcal{S}$ is the set of anchors and $\mathcal{G}$ has no edges unless two anchors happen to appear in each other’s pooled columns, so $\kappa = |\mathcal{S}|$ and the gauge is one constant per anchor.

Remark 7 (Downstream tasks are not gauge-invariant). Consider two anchors i, i' and a downstream decision that thresholds cosine similarity at a shared value θ , as in pair classification or a global retrieval cutoff. Offsets $a_i = +\Delta, a_{i'} = -\Delta$ leave $\mathcal{L}_{\text{diff}}$ unchanged by Proposition 5, yet move pairs of i above the threshold and pairs of i' below it, changing the decisions. Rank correlations computed across anchors, as in semantic textual similarity, are affected in the same way. The gauge is therefore task-relevant. A single offset shared across a component is the benign case: it shifts every cosine in the component equally, so it preserves all orderings and moves only the absolute location of θ .

B OBJECTIVE CONSTRAINTS AND IMPLEMENTATION NOTES

This appendix collects material deferred from Section 4: the constraint that fixes the student temperature at any term whose target is a transition row, and three details of the graph and walk construction.

B.1 TEMPERATURES TIED TO TRANSITION-ROW TARGETS

Section 4.4 gives each scale its own student temperature. Two terms have none to give: the sharpest graph scale and the row term both match transition rows, and Theorem 1 and Proposition 6 are statements about the student’s row at the graph’s own bandwidth. Matching at any other temperature would leave those results describing a different object than the one optimized. The proposition below records what goes wrong directly at the level of the loss.

Proposition 8 (Temperature ties). *Let Ω be a column set with $|\Omega| \geq 2$, and consider a term with target $\text{softmax}_\Omega(c_i^T/\tau)$ and student distribution $\text{softmax}_\Omega(c_i^S/\tau_S)$. (i) Over score assignments c_i^S the term has infimum 0, attained exactly on the affine family $c_{ij}^S = (\tau_S/\tau) c_{ij}^T + a_i$ on Ω — a shift family when $\tau_S = \tau$ and a rescaling otherwise. (ii) Suppose the ambient term at the same row also vanishes, on a set Ω_0 such that $\Omega \cap \Omega_0$ contains columns j, j' with $c_{ij}^T \neq c_{ij'}^T$. Then $\tau_S = \tau$.*

Proof. (i) If the divergence vanishes, the two softmax distributions coincide on Ω , and taking log-ratios of any two columns $j, j' \in \Omega$ (as in Appendix A.8) gives

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_S} = \frac{c_{ij}^T - c_{ij'}^T}{\tau}, \quad (26)$$

so $c_{ij}^S - (\tau_S/\tau) c_{ij}^T$ is constant on Ω ; call it a_i . Conversely, any score assignment of that form produces identical softmax distributions and zero divergence, so the infimum 0 is attained exactly on the stated family.

(ii) The ambient term carries the same temperature τ_0 on both sides, so its vanishing gives, by the shift case of (i), $c_{ij}^S = c_{ij}^T + b_i$ on Ω_0 . Combining with (i) on the overlap,

$$\left(1 - \frac{\tau_S}{\tau}\right) c_{ij}^T = a_i - b_i \quad \text{for every } j \in \Omega \cap \Omega_0, \quad (27)$$

and evaluating at two columns with $c_{ij}^T \neq c_{ij}^T$, forces $\tau_S = \tau$. $\square$

Part (ii) is the operative statement: with the ambient scale in the objective, a softmax-of-cosines target must be matched *at its own temperature*, or the two zero sets are disjoint and the objective acquires an artificial floor. The $r = 1$ target is the transition row P_i^T , a softmax of teacher cosines at the bandwidth τ_i , so $\tau_1(i) = \tau_i$; the walk targets are transition rows as well, so $\tau_w(j) = \tau_j$, which is the tie Proposition 6 assumes. The proposition is stated for one row and constrains only the *tie*, saying nothing about the value of τ , so it applies unchanged when every row carries its own bandwidth. The ambient scale uses one τ_0 on both sides, the same-temperature convention of knowledge distillation (Hinton et al., 2015), which by (i) makes its zero set exactly the shift family of Proposition 5.

B.2 THE REST OF THE TEMPERATURE LADDER

The tie fixes τ_1 ; the broader diffusion scales are then set by $\tau_r = \sqrt{r} \tau_1$, the $\sqrt{t}$ spread of a diffusion at nominal time r . We state this as a rule rather than derive it: the lazy walk takes $r/2$ real steps in expectation, so a strict derivation would carry a different constant, and at $r = 1$ the tie fixes the value regardless. What the rule buys is that the ladder has no free entries and inherits the per-row tie automatically — under entropic bandwidths $\tau_r(i) = \sqrt{r} \tau_i$ varies by row exactly as τ_1 does. The scale weights are set by the same principle in the other direction, $\omega_r = 1/r$ with $\omega_0 = \omega_1$, so a scale’s influence falls off with the diffusion time it describes.

B.3 TRUNCATION TOLERANCE

Sparse diffusion, the per-anchor candidate pool, and the walk support all perform the same operation: keep a subset of a probability row and renormalize. RIPPLE fixes one tolerance δ and keeps, per row, the smallest column set carrying $1 - \delta$ of the mass. Such a truncation perturbs the row by exactly δ in total variation and by $-\log(1 - \delta)$ nats in Kullback–Leibler divergence—a bound in the units of the loss itself—and by Lemma 3 the error is additive across steps, so an r -step target is within $r\delta$ of the exact one. Every stored array width is therefore a consequence of δ . Memory guards cap the widest rows; the build reports if either bound binds before the tolerance is met, in which case the guarantee does not hold.

B.4 NON-BACKTRACKING WALKS

The walk of Eq. (8) is non-backtracking: the step at $t + 1$ is drawn from $P_{j_t}^T$ conditioned on $j_{t+1} \neq j_{t-1}$, except at a node of degree one, where the edge just traversed is the only one available. Since rows are supervised with weight equal to their visit count, an excursion that returns immediately spends the walk budget re-supervising a row it has already seen; forbidding it raises the number of distinct rows a walk of given length reaches, and non-backtracking walks also have faster mixing and cover time on graphs of bounded degree (Alon et al., 2007). The constraint changes only the occupancy measure ν_B —which rows are chosen—and never a target: a visited node j is still matched against its own transition row P_j^T at τ_j . Proposition 6 is likewise unaffected, since it counts *traversed edges* and a repeated edge contributes no new constraint.

B.5 DENSE RATHER THAN SAMPLED WALK TARGETS

A natural alternative scores the sampled transitions themselves, $-\log p^S(j_{t+1} | j_t)$ summed along each path. The walk is first-order and each step is scored on the teacher’s own j_t , so that path likelihood factorizes into one-step terms whose expectation over the walk is exactly Eq. (9): the sampled form is an unbiased but higher-variance estimator of the dense one. Since P_j^T is already materialized during graph construction, we use the dense form and pay no variance. Nothing in the term couples two consecutive steps; it matches one-step transition rows at walk-visited nodes.